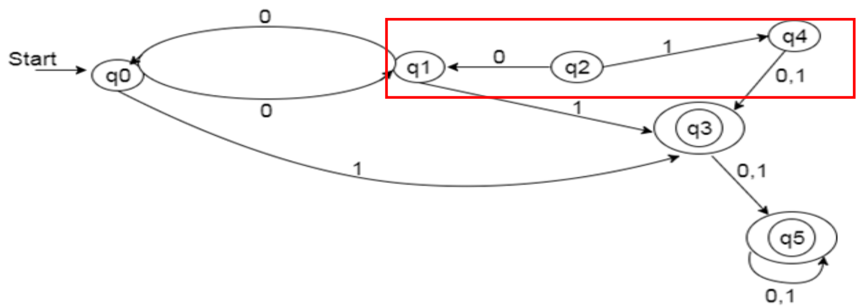
Minimization of DFA

Rules:

1. **Identify the unreachable state from final state and eliminate it.**



In the above diagram q2 and q4 is not reachable from start state.

2. **Now draw the table.**

states	0	1
q0	q1	q3
q1	q3	q0
q3	q5	q5
q5	q5	q5

3. **Separate the non-final and final states**

q0, q1	q3,q5

4. **Now compare the transition of all non-final states**

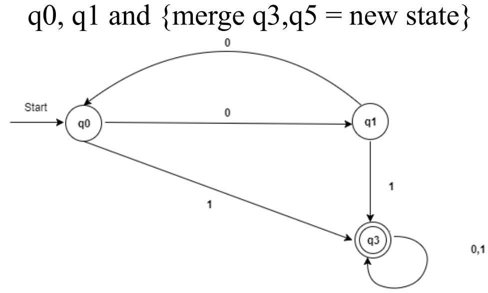
q0, q1

q0	q1	q3
q1	q3	q0

q0 and q1 is not same then separate the both state

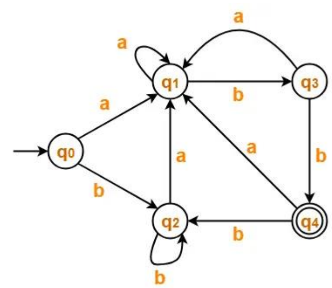
5. **After this the minimize DFA states are**

q0, q1 and {merge q3,q5 = new state}



Example 2

Convert the following DFA into Minimized DFA.



Step1: identify the non-reachable state.

All states are reachable from initial state, so no state elimination.

Step2: Draw the Transition table

states	a	b
q0	q1	q2
q1	q1	q3
q2	q1	q2
q3	q1	q4
q4	q1	q2

Step 3: separate the final and non-final state. [Important step]

Transitions	Non-Final states			Final State
	q0, q1, q2 , q3,			q4
Transition(q0,q1)= not same	q0, q2, q3	q1		q4
Transition(q0,q2)= same	q0, q2, q3	q1		q4
Transition(q0,q3)= not same	q0, q2,	q1	q3	q4
Transition(q1,q3)= not same	q0, q2,	q1	q3	q4

Step4: Final Minimized States

q0, q2,	q1	q3	q4

Step5: Draw the Machine